Rough Heston Variance at Zero: Negative Moments, Conditional Paths and a Test for the Zero Event

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Abstract

In the rough Heston model the instantaneous variance is zero with positive probability at every date. We show that, without drift, the variance also places so much probability just above zero that its negative moments on the event of positive variance are finite only below an explicit cutoff. The cutoff lies below one for every Hurst parameter and is about 0.26 when the Hurst parameter is 0.25, so the expected inverse variance is infinite even after the zero atom is excluded. The cutoff is the root of a Gamma-function equation that also governs how the variance falls to a zero. Conditional on zero variance at a given date, the expected variance path is an explicit mixture of powers of the time remaining, with positive weights and a leading power fixed by the same root. The zero event coincides with the finiteness of one weighted integral of past variance, whose conditional mean is known in closed form and is proportional to minus the log probability of the zero. Finally, two exact solutions driven by the same Brownian motion can separate only by changing sign from the first instant; whether they can separate at all remains open.

1 Question and answer

The rough Heston variance solves

$$V_t = v + \int_0^t K_\beta(t-s)(b-\kappa V_s)\,ds + \xi\int_0^t K_\beta(t-s)\sqrt{V_s}\,dW_s, \qquad K_\beta(t) = \frac{t^{\beta-1}}{\Gamma(\beta)}, \tag{1}$$

with $\beta = H + \frac{1}{2} \in (\frac{1}{2}, 1)$, $v, \xi > 0$ and $b, \kappa \geq 0$. Abi Jaber, Larsson and Pulido (2019) prove weak existence and uniqueness in law. Unlike the classical square-root process, the rough version reaches zero with positive probability at every fixed date, whatever the drift. Bourdon and Jeannin (2026, Theorem 3) establish this atom with a lower bound, and Friesen, Gerhold and Wiedermann (2026, Theorem 1.5) give it in closed form without mean reversion ($\kappa = 0$).

Once zero is an event of positive probability, a modeler needs to know what the variance looks like near it. How much probability sits just above zero, which decides whether $\mathbb{E}[V_t^{-1}; V_t > 0]$ is finite? How fast does the expected variance fall to a zero? And can two solutions driven by the same Brownian motion separate at a zero? The last question is pathwise uniqueness, and it is open. The available sufficient conditions for singular kernels require a diffusion coefficient that is Hölder of order above $1/(1+2H)$ (Mytnik and Salisbury, 2015; Prömel and Scheffels, 2025), the square root has order $1/2$, and the one proof claimed for rough Heston was withdrawn (Pannier and Jacquier, arXiv:1912.05917, because of an erroneous inequality in its Proposition 3.6).

We answer the first two questions exactly in the drift-free case $b = \kappa = 0$ and give one restriction on the third. Negative moments on $\{V_t > 0\}$ are finite exactly below a cutoff $\delta_\beta < 1$ (Theorem 1). Conditioning on $V_T = 0$ gives a linear equation for the conditional mean that can be solved in closed form (Theorem 2), with leading exponent $\rho_\beta = 2H(1+\delta_\beta)$. The solution yields exact formulas for weighted integrals of the past and an integral test for the zero event (Theorem 3). For the same-driver problem, any separation of exact solutions must oscillate in sign from its first instant (Proposition 1).

2 Setting and known facts

Write $L(t) = t^{-\beta}/\Gamma(1-\beta)$, so that $L * K_\beta = 1$, and $a = \xi^2/2$. For $u \geq 0$ let r_u be the nonnegative solution of the fractional Riccati equation $r_u = uK_\beta - K_\beta * (\kappa r_u + a r_u^2)$. The affine transform formula of Abi Jaber, Larsson and Pulido (2019) gives, for $b = \kappa = 0$,

$$\mathbb{E}e^{-uV_t} = \exp\{-v\,(L * r_u)(t)\}.$$

As $u \to \infty$, r_u increases to $r_\infty(t) = c_\beta t^{-\beta}$ (Friesen, Gerhold and Wiedermann, 2026, Theorem 3.2), where

$$c_\beta = \frac{2\Gamma(1-\beta)}{\xi^2\,|\Gamma(1-2\beta)|}, \qquad \lambda_\beta = \xi^2 c_\beta = \frac{2\Gamma(1-\beta)}{|\Gamma(1-2\beta)|}, \qquad \vartheta_\beta = \frac{c_\beta\,\Gamma(1-\beta)}{\Gamma(2-2\beta)}. \tag{2}$$

Since $(L * r_\infty)(t) = \vartheta_\beta t^{-2H}$, the atom is

$$p_t = \mathbb{P}(V_t = 0) = \exp\{-v\vartheta_\beta t^{-2H}\}. \tag{3}$$

Fractional integration of (1) and stochastic Fubini give, for all t almost surely,

$$(L * V)(t) = \frac{v\,t^{1-\beta}}{\Gamma(2-\beta)} + bt - \kappa\int_0^t V_s\,ds + \xi M_t, \qquad M_t = \int_0^t \sqrt{V_s}\,dW_s. \tag{4}$$

For a function f on $(0,\infty)$, integrable near zero, write $D^\beta f = (L * f)'$ for its Riemann-Liouville derivative where it exists. A direct calculation gives $D^\beta t^{-p} = -F_\beta(p)\,t^{-p-\beta}$ for $p < 1$, with

$$F_\beta(p) = -\frac{\Gamma(1-p)}{\Gamma(1-p-\beta)}, \qquad F_\beta(\beta) = \frac{\lambda_\beta}{2}.$$

3 Small positive variance

The atom fixes $\mathbb{P}(V_t = 0)$. How much probability lies in $(0, x]$ as $x \downarrow 0$ is a separate question, and it decides which negative moments $\mathbb{E}[V_t^{-q}; V_t > 0]$ are finite. Friesen, Gerhold and Wiedermann (2026, Theorem 1.2) prove that negative moments up to a time-dependent order are finite for kernels with finite $K(0)$, a class that excludes the rough kernel. Let p_β be the unique solution in $(\beta, 1)$ of $F_\beta(p_\beta) = \lambda_\beta$, and set

$$\delta_\beta = \frac{p_\beta - \beta}{2H}.$$

Theorem 1. *Let $b = \kappa = 0$, $v > 0$ and $t > 0$. Then $\mathbb{E}[V_t^{-q}; V_t > 0] < \infty$ if and only if $q < \delta_\beta$, and $\log \mathbb{P}(0 < V_t \leq x)/\log x \to \delta_\beta$ as $x \downarrow 0$. The cutoff satisfies $0 < \delta_\beta < 1$ for every $H \in (0, \frac{1}{2})$, so $\mathbb{E}[V_t^{-1}; V_t > 0] = \infty$.*

The proof compares the gap $d = r_\infty - r_1$ between the maximal and the unit Riccati solutions with powers of t, using a first-contact property of D^β.

Lemma 1. *Let f be continuous on $(0,\infty)$, integrable near zero, and continuously differentiable near some $t_* > 0$, with $f > 0$ on $(0, t_*)$ and $f(t_*) = 0$. Then*

$$D^\beta f(t_*) = -\frac{\beta}{\Gamma(1-\beta)}\int_0^{t_*} \frac{f(s)}{(t_* - s)^{1+\beta}}\,ds < 0.$$

Proof. Fix $\eta \in (0, t_*)$ with f continuously differentiable near $[t_* - \eta, t_*]$ and write $(L * f)(t) = \int_0^{t_*-\eta} L(t-s)f(s)\,ds + \int_0^{t-t_*+\eta} L(u)f(t-u)\,du$ for t near t_*. Differentiating at t_* gives $\int_0^{t_*-\eta} L'(t_* - s)f(s)\,ds + L(\eta)f(t_* - \eta) + \int_0^\eta L(u)f'(t_* - u)\,du$. Integrating the last term by parts, with $L(u)f(t_* - u) = O(u^{1-\beta}) \to 0$ as $u \downarrow 0$, cancels the middle term and leaves $\int_0^{t_*} L'(t_* - s)f(s)\,ds$, where $L'(x) = -\beta x^{-1-\beta}/\Gamma(1-\beta)$. The integrand is $O((t_* - s)^{-\beta})$ near t_*. □

Lemma 2. *For every $p \in (\beta, p_\beta)$ there is C_p, and there is $c > 0$, such that $c\, s^{-p_\beta} \le d(s) \le C_p s^{-p}$ for all $s \ge 1$.*

Proof. Five facts are used. (a) Applying $L*$ to the Riccati equation gives $(L * r_1)(t) = 1 - a\int_0^t r_1^2$, and $(L*r_\infty)(t) = \vartheta_\beta t^{-2H}$, so $D^\beta d + Qd = 0$ on $(0,\infty)$ with $Q = a(r_\infty + r_1)$. (b) Since $0 \le r_1 \le r_\infty$ and $2ac_\beta = \lambda_\beta$, $t^\beta Q(t) \le \lambda_\beta$; by scaling, $t^\beta r_1(t) = r_{t^{2H}}(1) \to c_\beta$, so $t^\beta Q(t) \to \lambda_\beta$. (c) $r_1 \le K_\beta$, and $K_\beta(t) = o(t^{-\beta})$ as $t \downarrow 0$, so $d(t) \sim c_\beta t^{-\beta}$ there. (d) r_1 is continuously differentiable on $(0,\infty)$. The equation makes r_1 locally Hölder of order β on $(0,\infty)$, hence also r_1^2, and the fractional integral of order β maps locally β-Hölder functions to C^1 functions because $2\beta > 1$ (Samko, Kilbas and Marichev, 1993). (e) $d > 0$ on $(0,\infty)$, since at a first zero Lemma 1 would give $D^\beta d < 0$ while (a) gives $D^\beta d = 0$.

Upper bound. Since F_β is continuous, $F_\beta(\beta) = \lambda_\beta/2$ and p_β is the only solution of $F_\beta = \lambda_\beta$ in $(\beta, 1)$, $F_\beta(p) < \lambda_\beta$, so by (b) there is $t_0 \ge 1$ with $t^\beta Q(t) > F_\beta(p)$ for $t \ge t_0$. By (c) and $p > \beta$, there is C with $g = Ct^{-p} - d > 0$ on $(0, t_0]$. If g had a zero, its first zero $t_* > t_0$ would satisfy, using $d(t_*) = Ct_*^{-p}$,

$$D^\beta g(t_*) = -CF_\beta(p)t_*^{-p-\beta} + Q(t_*)d(t_*) = C\big[t_*^\beta Q(t_*) - F_\beta(p)\big]t_*^{-p-\beta} > 0,$$

contradicting Lemma 1.

Lower bound. Let $w(s) = 1 + M(1-s)$ for $s < 1$ and $w(s) = s^{-p_\beta}$ for $s \ge 1$, with $M > 2p_\beta/(1-p_\beta)$, and let $h = w - s^{-p_\beta}$ on $(0,1)$, zero elsewhere. Then $\int_0^1 h = 1 + M/2 - 1/(1-p_\beta) > 0$. For $t > 1$,

$$(D^\beta + Q)w(t) = \big[t^\beta Q(t) - \lambda_\beta\big]t^{-p_\beta-\beta} - \frac{\beta}{\Gamma(1-\beta)}\int_0^1 \frac{h(s)}{(t-s)^{1+\beta}}\,ds.$$

The first term is nonpositive by (b), and $t^{1+\beta}$ times the integral tends to $\int_0^1 h > 0$, so there is $t_1 > 1$ with $(D^\beta + Q)w < 0$ on $[t_1,\infty)$. Since w is bounded on $(0,1]$ and d is positive and diverges at zero, there is $c > 0$ with $g = d - cw > 0$ on $(0, t_1]$. At a first zero $t_* > t_1$, $D^\beta g(t_*) = -c\,(D^\beta + Q)w(t_*) > 0$, again contradicting Lemma 1. Hence $d \ge cs^{-p_\beta}$ on $[t_1,\infty)$, and continuity and positivity extend the bound to $[1, t_1]$ after reducing c. □

Proof of Theorem 1. Write $G(u) = \mathbb{E}[e^{-uV_t}; V_t > 0] = \mathbb{E}e^{-uV_t} - p_t$. By scaling, $r_u(s) = \ell^\beta r_1(\ell s)$ with $\ell = u^{1/(2H)}$, so $(L * r_u)(t) = \ell^{2H}(L * r_1)(\ell t)$ and

$$G(u) = p_t\Big(\exp\big\{v\,\ell^{2H}(L*d)(\ell t)\big\} - 1\Big).$$

For $s \ge 2$, Lemma 2 and $d \le r_\infty$ on $(0,1)$ give $(L*d)(s) \le C(s^{-\beta} + s^{1-\beta-p})$ for each $p \in (\beta, p_\beta)$, and $(L*d)(s) \ge c\int_1^{s/2} L(s-r)r^{-p_\beta}dr \ge c's^{1-\beta-p_\beta}$. Since $1-\beta-p > -\beta$ and $\ell^{2H}\ell^{1-\beta-p} = \ell^{\beta-p} = u^{-(p-\beta)/(2H)}$, it follows that $G(u) \ge c\,u^{-\delta_\beta}$ and $G(u) \le C_\epsilon u^{-\delta_\beta+\epsilon}$ for every $\epsilon > 0$ and large u. The moment statement follows from $\mathbb{E}[V_t^{-q}; V_t > 0] = \Gamma(q)^{-1}\int_0^\infty u^{q-1}G(u)\,du$, which diverges at $q = \delta_\beta$ like $\int^\infty du/u$. The probability statement follows from $\mathbb{P}(0 < V_t \le x) \le e\,G(1/x)$ and from $G(u) \le \mathbb{P}(0 < V_t \le y) + e^{-uy}$ with $y = M\log u/u$ and $M > \delta_\beta$.

It remains to show $0 < \delta_\beta < 1$. First, $\delta_\beta > 0$ because $p_\beta > \beta$. With $\rho_\beta = p_\beta + \beta - 1$, the Gamma recurrence gives $F_\beta(p) = r\Gamma(\beta - r)/\Gamma(1-r)$ for $r = p + \beta - 1$, so ρ_β solves $\rho\Gamma(\beta-\rho)/\Gamma(1-\rho) = \lambda_\beta$, and $\delta_\beta < 1$ is equivalent to $\rho_\beta < 4H$. If $H > 1/6$, then $4H > \beta > \rho_\beta$. If $H \le 1/6$, the function Ψ of (5) below is increasing on $(-1,\infty)$ with its only zero there at $-\rho_\beta$ (proof of Theorem 2), so the claim is $\Psi(-4H) < 0$. At $H = 1/6$, $1/\Gamma(\beta - 4H) = 0$ and $\Psi(-4H) = -4H < 0$. For $H < 1/6$, using $\lambda_\beta = 4H\,\Gamma(\frac12 - H)/\Gamma(1-2H)$, the claim is

$$\log\Gamma(\tfrac12 - H) + \log\Gamma(1-4H) < \log\Gamma(1-2H) + \log\Gamma(\tfrac12 - 3H).$$

Both pairs of arguments sum to $\frac32 - 5H$, and the larger argument on the right, $1-2H$, exceeds both arguments on the left, so the inequality follows from strict convexity of $\log\Gamma$. □

The cutoff is small for rough paths of moderate roughness and approaches one only as $H \to 0$ (Table 1). At $H = 0.25$, $\mathbb{E}[V_t^{-q}; V_t > 0]$ is already infinite for $q \ge 0.258$.

Table 1: Boundary exponents. δ_β is the negative-moment cutoff (Theorem 1); $\rho_\beta = 2H(1+\delta_\beta)$ and A_0 describe the conditional mean before a zero (Theorem 2).

H	β	p_β	δ_β	ρ_β	A_0
0.10	0.60	0.7174	0.5868	0.3174	0.5619
0.25	0.75	0.8790	0.2580	0.6290	0.2131
0.40	0.90	0.9626	0.0783	0.8626	0.0557

4 The history of a terminal zero

Fix $T > 0$, set $b = \kappa = 0$, and let $\mathbb{Q}_T = \mathbb{P}(\,\cdot \mid V_T = 0)$ and $m_T(s) = \mathbb{E}_{\mathbb{Q}_T} V_s$. Define

$$\Psi(z) = z + \lambda_\beta \frac{\Gamma(z+1)}{\Gamma(z+\beta)}. \tag{5}$$

Theorem 2. *(i) Under $\mathbb{Q}_T$, the process $W_s + \xi c_\beta \int_0^s (T-r)^{-\beta}\sqrt{V_r}\,dr$, $s < T$, is a Brownian motion, so the conditioned variance carries the additional drift $-\lambda_\beta \int_0^s K_\beta(s-r)(T-r)^{-\beta} V_r\,dr$.*

(ii) Ψ has exactly one zero $-\rho_n$ in each interval $(-n-\beta, -n)$, $n \geq 0$, and no other real zeros. The numbers $A_n = 1/\Psi'(-\rho_n)$ are positive and sum to one, and

$$m_T(s) = v \sum_{n\geq 0} A_n \Big(1 - \frac{s}{T}\Big)^{\rho_n}, \qquad 0 \leq s \leq T. \tag{6}$$

The leading exponent is $\rho_0 = \rho_\beta = 2H(1+\delta_\beta) \in (2H, \beta)$, the second is $\rho_1 = 2\beta$, and $A_0 = [1 + \rho_\beta\{\psi(1-\rho_\beta) - \psi(\beta - \rho_\beta)\}]^{-1}$ with $\psi = \Gamma'/\Gamma$. In particular $\mathbb{E}[V_{T-h} \mid V_T = 0] = vA_0(h/T)^{\rho_\beta} + O(h^{2\beta})$ as $h \downarrow 0$.

Proof. (i) For $u > 0$ let $Z_s^u = \mathbb{E}[e^{-uV_T} \mid \mathcal{F}_s]/\mathbb{E}e^{-uV_T}$. The affine transform formula (Abi Jaber, Larsson and Pulido, 2019, Section 4) gives $dZ_s^u = -\xi r_u(T-s)\sqrt{V_s}\,Z_s^u\,dW_s$. Since $\mathbb{E}e^{-uV_T} \geq p_T$, $Z^u \leq 1/p_T$, and $Z_s^u \to Z_s = \mathbb{P}(V_T = 0 \mid \mathcal{F}_s)/p_T$ as $u \to \infty$. On $[0, T-\varepsilon]$, $0 \leq r_u(T-s) \uparrow c_\beta(T-s)^{-\beta} \leq c_\beta \varepsilon^{-\beta}$ and $\mathbb{E}V_s = v$, so the integrands converge in $L^2(ds\,d\mathbb{P})$ by dominated convergence and $dZ_s = -\xi c_\beta (T-s)^{-\beta}\sqrt{V_s} Z_s\,dW_s$. The limit is a strictly positive stochastic exponential on each $[0, T-\varepsilon]$, and Girsanov's theorem gives the stated drift.

(ii) Under $\mathbb{Q}_T$, $V_s = v - \lambda_\beta \int_0^s K_\beta(s-r)(T-r)^{-\beta} V_r\,dr + \xi \int_0^s K_\beta(s-r)\sqrt{V_r}\,d\widetilde{W}_r$. For fixed $s < T$, the process $u \mapsto \int_0^u K_\beta(s-r)\sqrt{V_r}\,d\widetilde{W}_r$, $u \leq s$, is a square-integrable $\mathbb{Q}_T$-martingale, with second moment at most $(v/p_T)\int_0^s K_\beta(s-r)^2 dr$ because $\mathbb{E}_{\mathbb{Q}_T} V_r \leq \mathbb{E}V_r/p_T = v/p_T$. Its value at $u = s$ has mean zero, so

$$m_T(s) = v - \lambda_\beta \int_0^s K_\beta(s-r)(T-r)^{-\beta} m_T(r)\,dr. \tag{7}$$

Put $T = 1$ (the general case follows by scaling), $y = -\log(1-s)$ and $\varphi(y) = m_T(s)/v$. The substitution $r = 1 - e^{-y'}$ turns (7) into the renewal equation $\varphi = 1 - \lambda_\beta\, k * \varphi$ with $k(y) = (1-e^{-y})^{\beta-1}/\Gamma(\beta)$, which has a unique locally bounded solution. Since $\int_0^\infty e^{-zy}k(y)\,dy = \Gamma(z)/\Gamma(z+\beta)$, the Laplace transform of φ is $1/\Psi(z)$. Expanding $(1-e^{-y})^{\beta-1} = \sum_{n\geq 0}\gamma_n e^{-ny}$ with $\gamma_n = (1-\beta)_n/n! > 0$ gives

$$\Psi(z) = z + \frac{\lambda_\beta}{\Gamma(\beta)}\Big(\gamma_0 + \sum_{n\geq 1}\gamma_n \frac{z}{z+n}\Big),$$

a convergent sum because $\gamma_n = O(n^{-\beta})$. Hence $\Psi' = 1 + \lambda_\beta\Gamma(\beta)^{-1}\sum_{n\geq 1}\gamma_n n/(z+n)^2 > 0$ on the real line away from the poles $-1, -2, \ldots$, and on each interval between consecutive poles, and on $(-1, \infty)$, Ψ increases from $-\infty$ to $+\infty$. It therefore has exactly one zero in each such interval,

with $\Psi' > 0$ there. Since $\Gamma(z+1)/\Gamma(z+\beta)$ vanishes at $z = -n-\beta$, $\Psi(-n-\beta) = -n-\beta < 0$, which places the zero in $(-n-\beta, -n)$.

For the expansion of $1/\Psi$ we adapt the interlacing method of Kuznetsov, Kyprianou and Pardo (2012) and truncate the sum at $n = N$. The truncated function Ψ_N is rational, increasing between its poles, and of order z at infinity, so $1/\Psi_N(z) = \sum_{n=0}^{N} A_n^N/(z+\rho_n^N)$ with $\rho_n^N > n$, $A_n^N > 0$ and $\sum_n A_n^N = 1$. As $N \to \infty$, $\Psi_N \to \Psi$ locally uniformly away from the poles, so each $\rho_n^N \to \rho_n$ and $A_n^N \to A_n = 1/\Psi'(-\rho_n)$. For $x > 0$ the tail $\sum_{n \geq m} A_n^N/(x+\rho_n^N)$ is at most $1/(x+m)$, uniformly in N, which gives $1/\Psi(x) = \sum_n A_n/(x+\rho_n)$ for $x > 0$ and, by analytic continuation, for $x > -\rho_0$. Monotone convergence and $x/\Psi(x) \to 1$ give $\sum_n A_n = 1$. The Laplace transform of $\sum_n A_n e^{-\rho_n y}$ is therefore $1/\Psi$, and uniqueness of the Laplace transform gives (6). The equation for ρ_0 is $\Psi(-\rho) = 0$, which is the equation for ρ_β in the proof of Theorem 1. Direct substitution, using $\lambda_\beta \Gamma(1-2\beta) = -2\Gamma(1-\beta)$ and $\Gamma(-\beta) = -\Gamma(1-\beta)/\beta$, gives $\Psi(-2\beta) = 0$, and differentiating (5) at $-\rho_0$ gives A_0. Finally $\rho_0 > 2H$ because $\Psi(-2H) = 2H > 0$ (proof of Theorem 3). □

The conditional mean is therefore a positive mixture of powers whose leading term vanishes like $(T-s)^{\rho_\beta}$, faster than $(T-s)^{2H}$. Its exponent and the negative-moment cutoff are tied by $\rho_\beta = 2H(1+\delta_\beta)$, so one Gamma-function root controls both the mass just above zero at a fixed date and the rate at which the expected variance falls to a zero.

Theorem 3. *Let $b = \kappa = 0$ and $J_q(T) = \int_0^T V_s (T-s)^{-q-1}\, ds$. For $0 < q < \rho_\beta$,*

$$\mathbb{E}[J_q(T) \mid V_T = 0] = \frac{v\, T^{-q}}{\Psi(-q)} = \frac{v\, T^{-q}}{\lambda_\beta \Gamma(1-q)/\Gamma(\beta-q) - q}, \tag{8}$$

and the conditional expectation is infinite for $q \geq \rho_\beta$. In particular $\mathbb{E}[J_{2H}(T) \mid V_T = 0] = vT^{-2H}/(2H)$, and

$$\{V_T = 0\} = \{J_{2H}(T) < \infty\} \qquad \textit{almost surely.} \tag{9}$$

Proof. By Tonelli and (6), $\mathbb{E}_{\mathbb{Q}_T} J_q(T) = vT^{-q} \sum_n A_n \int_0^1 (1-x)^{\rho_n - q - 1} dx$. For $q \geq \rho_0$ the term $n = 0$ is infinite. For $q < \rho_0$ every term is finite, the sum is $vT^{-q} \sum_n A_n/(\rho_n - q)$, and this equals $vT^{-q}/\Psi(-q)$ by the partial-fraction expansion at $x = -q > -\rho_0$. At $q = 2H = 2\beta - 1$, $\Gamma(\beta - q) = \Gamma(1-\beta)$ and $\Gamma(1-q) = (1-2\beta)\Gamma(1-2\beta)$, so $\lambda_\beta \Gamma(1-q)/\Gamma(\beta-q) = 4H$ and $\Psi(-2H) = 2H$. For (9), $J_{2H}(T)$ has finite $\mathbb{Q}_T$-mean, so it is finite almost surely on $\{V_T = 0\}$. On $\{V_T > 0\}$, continuity keeps V above a positive level near T, and $\int^T (T-s)^{-2H-1} ds = \infty$. □

By continuity V_T is already determined by the path before T; the content of (9) is that one explicit weighted integral decides the event.

Remark. The constant $1/(2H)$ can be checked against the atom (3). Under $\mathbb{Q}_T$ the Brownian drift in Theorem 2(i) has accumulated square $\xi^2 c_\beta^2 J_{2H}(T)$, which has finite mean by Theorem 3. Hence $\log Z_s = \frac{1}{2}\xi^2 c_\beta^2 \int_0^s (T-r)^{-2\beta} V_r\, dr$ plus a $\mathbb{Q}_T$-martingale that is square-integrable up to T. Since $\{V_T = 0\} \in \mathcal{F}_{T-}$, $Z_s \to 1/p_T$ $\mathbb{Q}_T$-almost surely as $s \uparrow T$, and taking expectations gives $\frac{1}{2}\xi^2 c_\beta^2\, \mathbb{E}_{\mathbb{Q}_T} J_{2H}(T) = -\log p_T = v\vartheta_\beta T^{-2H}$, which matches Theorem 3 because $\xi^2 c_\beta^2/(4H) = \vartheta_\beta$. This is an instance of the identity between relative entropy and the energy of the Girsanov drift (Lehec, 2013), since the relative entropy of $\mathbb{Q}_T$ with respect to $\mathbb{P}$, namely $-\log p_T$, equals half the expected accumulated squared drift.

5 Same-driver solutions must oscillate

Proposition 1. *Let V and $\widetilde{V}$ be continuous nonnegative adapted solutions of (1) on one filtered probability space satisfying the usual conditions, with the same Brownian motion W and the same parameters (v, b, κ, ξ), and let $D = V - \widetilde{V}$ and $\sigma = \inf\{t : D_t \neq 0\}$. Almost surely on $\{\sigma < \infty\}$, every interval $(\sigma, \sigma+\varepsilon)$ contains times with $D > 0$ and times with $D < 0$.*

Proof. By uniqueness in law (Abi Jaber, Larsson and Pulido, 2019), $\widetilde{V}$ has the law of V, so $\mathbb{E}\int_0^t (V_s + \widetilde{V}_s)\,ds < \infty$ for every t. Both sides of (4) are continuous in t, so it holds for V and $\widetilde{V}$ simultaneously at all t outside one null set. There, $(L * D)(t) = -\kappa \int_0^t D_s\,ds + \xi N_t$ with $N = \int(\sqrt{V} - \sqrt{\widetilde{V}})\,dW$, a square-integrable martingale on bounded intervals. Fix an integer T and let $\tau = \inf\{t : D_t < 0\} \wedge T$. On $[0, \tau]$, $D \geq 0$, so $(L * D)(\tau) \geq 0$ because $L > 0$, while $\mathbb{E}(L * D)(\tau) = -\kappa \mathbb{E}\int_0^\tau D_s\,ds \leq 0$ by optional stopping. Hence $(L * D)(\tau) = 0$ almost surely, and $D = 0$ on $[0, \tau]$. The same argument applied to $-D$ shows that $D = 0$ up to the first time it is positive, capped at T. Intersecting these full-probability events over integer T gives $\inf\{t : D_t > 0\} = \inf\{t : D_t < 0\} = \sigma$ almost surely on $\{\sigma < \infty\}$, which is the claim. □

The proposition concerns exact solutions in continuous time. It leaves pathwise uniqueness open and constrains the answer, since a construction of two distinct solutions driven by one Brownian motion must produce a difference that changes sign in every right neighborhood of the first separation. It says nothing directly about discretized schemes, whose paths are not solutions of (1).

6 Why it matters

The rough Heston variance hits zero with positive probability, and simulation schemes and pricing formulas have to deal with that boundary. In the drift-free model the boundary is now described exactly. Expectations of quantities divided by the variance must be taken on the event that the variance is positive, and even there the first negative moment is infinite for every Hurst parameter, with negative moments finite only below the cutoff in Table 1. The same root fixes the expected path into a zero, and the zero event can be read off one weighted integral of past variance whose conditional mean is tied to the probability of the zero.

Notes. All constants and Table 1 were checked numerically at 30 digits, and $0 < \delta_\beta < 1$ was also checked on a grid of H from 0.001 to 0.499 (the largest value, $\delta_\beta = 0.99447$, is at $H = 0.001$). At $\beta \in \{0.6, 0.75, 0.9\}$ the first 80 terms of (6) solve (7) to within 3.1×10^{-4}; at $\beta = 0.75$ they carry total weight 0.9757, the rest of the weight sitting on powers of order 80 and above. In the papers we checked we found no earlier statement of the negative-moment cutoff, the conditional power mixture, the weighted-integral test or the oscillation result, while the terminal tilt of affine processes (Abi Jaber, Larsson and Pulido, 2019) and the entropy identity (Lehec, 2013) have clear antecedents. The search does not establish priority.

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